Lesson 5.7 Guided Notes

Session 1 · Surface Area of Rectangular Prisms

I can find the surface area of a rectangular prism by adding the areas of its six faces. · 6.G.A.4

Word / Spanish	Meaning — complete the blank	Example
surface area / área superficial	The total _____ of all the faces of a solid.	add every face
congruent / congruente	Exactly the same _____ and shape.	opposite faces match
net / plantilla	A flat _____ that folds into a solid.	shows all 6 faces

Key idea

A rectangular prism has _____ pairs of matching faces.

SA = 2lw + 2lh + _____.

Surface area is measured in _____ units.

Worked example

A closed box is 6 cm long, 4 cm wide, and 3 cm tall. Find its surface area.

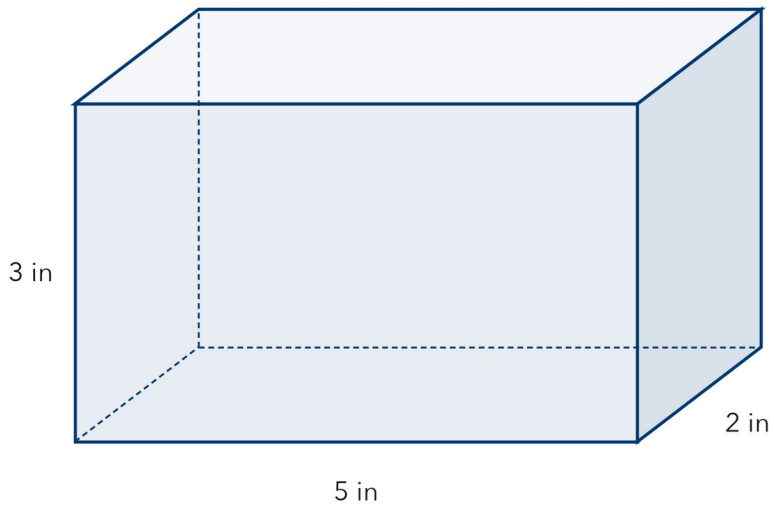
Top: 6 × 4 = _____ cm². Front: 6 × 3 = 18 cm². End: 4 × 3 = _____ cm².

Double each: 48 + 36 + 24.

Answer: SA = _____ cm²

You try

1. Find the surface area of the prism.



_____

2. Find the surface area of each cube.
a. edges 2 m _____ m²
b. edges 10 in _____ in²

_____

Sum it up: Surface area is different from volume because ___.

_____

Session 2 · Surface Area of Triangular Prisms

I can find the surface area of a triangular prism using its net. · 6.G.A.4

Word / Spanish	Meaning — complete the blank	Example
triangular prism / prisma triangular	A solid with 2 triangle _____ and 3 rectangle faces.	a tent shape
lateral face / cara lateral	A side face that is _____ a base.	the 3 rectangles
surface area / área superficial	The total area of all the _____.	add all 5 faces

Key idea

Add the areas of all _____ faces: 2 triangles and 3 rectangles.

Each triangle uses ½ × base × _____.

Each rectangle is one triangle _____ × the prism's length.

Worked example

A prism is 6 ft long. Each triangle end has a base of 8 ft, a height of 3 ft, and two 5 ft sides. Find its surface area.

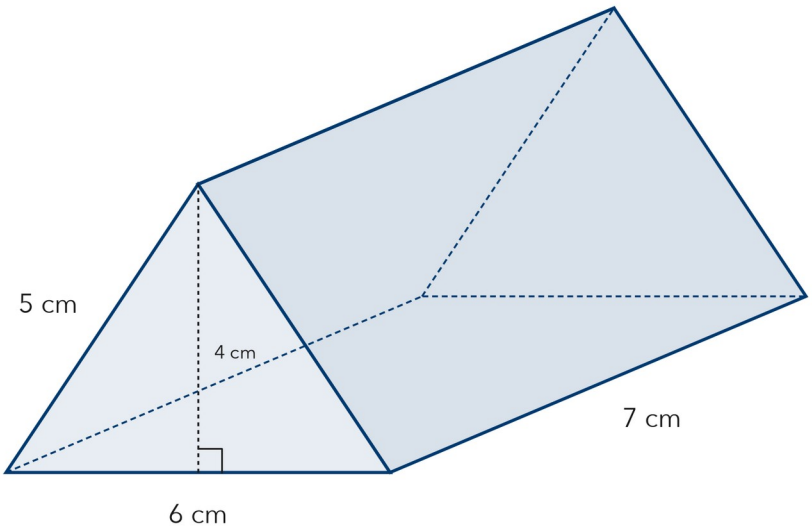
One end: ½ × 8 × 3 = _____ ft². Both ends: 24 ft².

Rectangles: 8 × 6 + 5 × 6 + 5 × 6 = _____ ft².

Answer: SA = 24 + 108 = _____ ft²

You try

1. Find the surface area of the triangular prism.



_____

2. Find each area.
a. 2 triangles, base 10 in, height 2 in _____ in²
b. rectangle 4 in by 9 in _____ in²

_____

Sum it up: For a triangular prism, I use the triangle's height for ___ and its sides for ___.

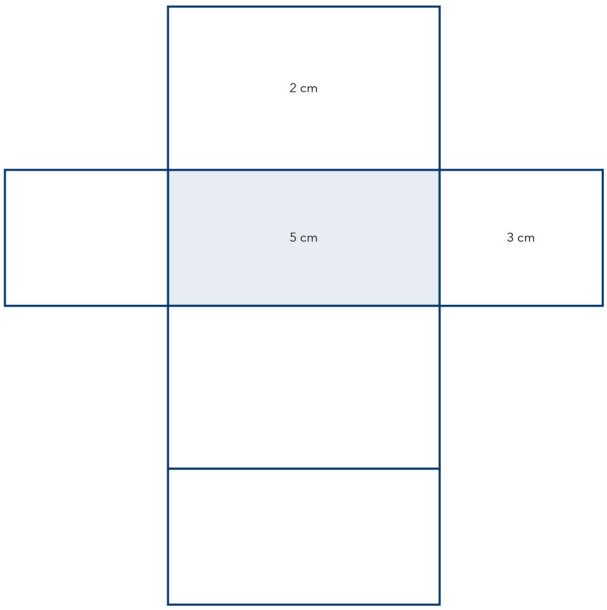
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Session 1 · Model, reason, and explain

Surface Area of Rectangular Prisms · 6.G.A.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Use a labeled net of a closed 5-by-3-by-2 cm prism to find surface area. Then remove the top 5-by-3 face and find exposed area of the open box.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: includes a missing top or omits one of a matching pair.. Explain what changes and what stays true.

_____

_____

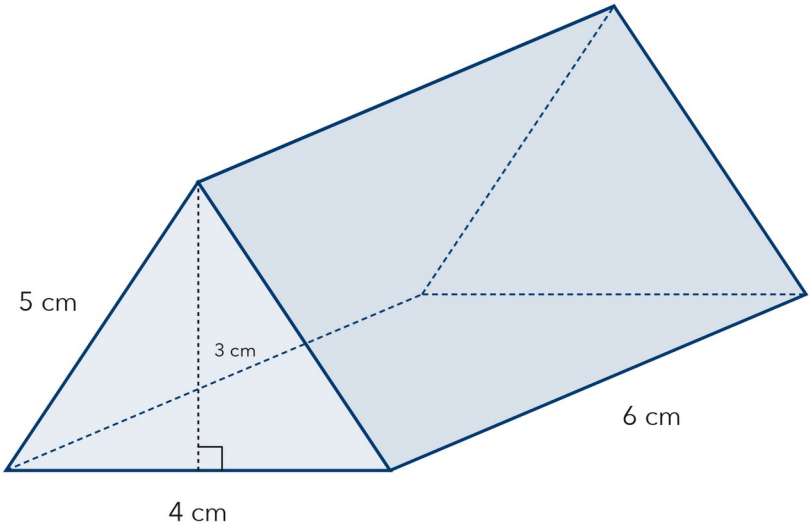
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Surface Area of Triangular Prisms · 6.G.A.4

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A triangular prism has 3-4-5 right-triangle ends and length 6 cm. Label the triangle height separately from prism length, and find surface area.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: uses prism length as triangle height or slanted triangle side as height.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?